

David Mekhtiev

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EDUCATION

University of Pennsylvania

Master of Science in Engineering, Computer Science

Philadelphia, PA

Jan 2027 — Expected 2028

- Incoming for the Spring 2027 semester with a focus on software systems. This is an online program.

University of Michigan

Bachelor of Science in Engineering, Computer Science and Engineering

Ann Arbor, MI

Aug 2021 — May 2025

- GPA: 3.84 / 4.0 | Dean's List, Engineering Scholarship of Honor

WORK EXPERIENCE

Hardware & Systems Engineer

Sep 2025 — Present

Analog Devices

Wilmington, MA

- Built and maintain a full-stack workflow automation platform used by ~50 engineers in the digital IC design group, cutting the longest turnarounds from days to a few hours.
- Developed the React + TypeScript frontend and a Python FastAPI REST backend backed by PostgreSQL, handling ~100-200 jobs per day.
- Implemented authentication and role-based authorization via SSO, restricting access to our Linux HPC cluster and licensed software.
- Integrated an existing job scheduler that distributes work and resolves inter-job dependencies across our HPC Linux cluster.

Hardware & Systems Engineer Intern

May 2025 — Aug 2025

Analog Devices

Wilmington, MA

- Built an ML model, plus the data generation, augmentation, training, and evaluation pipeline around it, to predict performance and power usage for a proprietary audio/low-power ML hardware subsystem used in 2 product lines.
- Reduced the time to obtain performance and power estimates by several orders of magnitude (hours to seconds).

Hardware & Systems Engineer Intern

May 2024 — Aug 2024

Analog Devices

Wilmington, MA

- Automated an existing workflow for a Neural Processing Unit's SDK for PPA analysis. This involved modifying memory map tables, generating linker scripts, compiling TFLite neural networks into binary executables, running SystemC simulations, scraping simulation logs for performance data, generating/simulating parameterizable RTL, and performing power usage analysis.

Teaching Assistant, EECS 483 (Compiler Design)

Jan 2025 — May 2025

University of Michigan

Ann Arbor, MI

- Led a weekly discussion section, held office hours, and co-designed and graded exams for Professor [Max New](#)'s course; earned a 4.9/5 average student rating during the [winter 2025 semester](#).

PROJECTS

Sharded Paxos Key/Value Service

Nov 2024 — Dec 2024

- Built a fault-tolerant, sharded key/value store in Go, using Paxos to keep replica groups consistent through failures, with a shard master that rebalances and migrates shards as groups join and leave.

Parametrizable Superscalar Out-of-Order Processor

Jan 2024 — Apr 2024

- Designed a synthesizable 32-bit RISC-V out-of-order core in SystemVerilog (MIPS R10K-inspired microarchitecture) with a tournament branch predictor, load-store queue, instruction prefetching, and multi-ported non-blocking caches.
- Built a visual React debugger that parses FSDb waveform dumps to visualize every CPU component's signals cycle by cycle.

Virtual Memory Pager

Oct 2023 — Nov 2023

- Implemented the virtual memory pager of a Unix-like kernel in C++, supporting `fork()`, address space creation, copying, destruction, and growth, and context switching between processes.

SKILLS

- **Languages:** C, C++, CUDA, Go, JavaScript, Python, Rust, SQL, SystemVerilog, TypeScript
- **Frameworks & Libraries:** Boost, Electron, FastAPI, FlatBuffers, Matplotlib, NumPy, React, TensorFlow
- **Tools & Infrastructure:** Docker, GDB, Git, Linux, LLDB, PostgreSQL, Valgrind